

Skills Summary

Radicals and Rational Exponents — The Illegal Split

Use this reference while completing your worksheet or when studying.

Skill 1 — Roots of a sum: test before you split

Legal: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$ — a radical distributes over multiplication and division.

Not legal: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$, and in exponent form $(a+b)^{1/2} \neq a^{1/2} + b^{1/2}$. The radical sign is a grouping symbol: finish the addition underneath it, then take one root of the single number you get.

Why: squaring $\sqrt{a} + \sqrt{b}$ gives $a + b + 2\sqrt{ab}$, not $a + b$. The split silently drops the cross term $2\sqrt{ab}$, so it is wrong for every pair of positive numbers.

Example: Compute $\sqrt{49 + 576}$, and test the split against it.

Step 1. The radicand is a sum, so the product rule does not apply — add first, then take one root, and use the split only to see how far off it lands.

Step 2. $49 + 576 = 625$, and $\sqrt{625} = 25$; the split would give $\sqrt{49} + \sqrt{576} = 7 + 24 = 31$. $\sqrt{49 + 576} = 25$, not 31.

Try it: Compute $\sqrt{9 + 40}$.

check: 7

Skill 2 — Simplify each radical, then combine

Rule: pull perfect-square factors out one radical at a time: $\sqrt{k^2m} = k\sqrt{m}$. Two radicals may be ADDED only when they are *like radicals* — same number under the root — and then they combine like terms: $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.

Not legal: pushing two radicals back into one, $\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$. That is the same illegal split run backwards.

Example: Simplify $\sqrt{75} + \sqrt{12}$.

Step 1. Two unlike radicals cannot be added as they stand and must not be merged into one root, so simplify each radicand by its largest square factor first.

Step 2. $\sqrt{75} = \sqrt{25 \cdot 3} = 5\sqrt{3}$ and $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$; both are now multiples of $\sqrt{3}$, so $5\sqrt{3} + 2\sqrt{3}$ combines like $5x + 2x$. $\sqrt{75} + \sqrt{12} = 7\sqrt{3}$

Try it: Simplify $\sqrt{32} + \sqrt{8}$.

check: $6\sqrt{2}$

Skill 3 — Radical equations: square a whole side

Method: get the radical alone on one side, then square that *entire* side — never the terms inside it one at a time. Squaring undoes the root: $(\sqrt{u})^2 = u$.

Then check. Squaring can create an answer that fails the original equation, so every solution goes back into the equation you started with.

Example: Solve $\sqrt{x+7} = 4$.

Step 1. The radical already stands alone, so squaring the whole side is the tool that removes it — splitting the radicand into $\sqrt{x} + \sqrt{7}$ would change the equation before solving it.

Step 2. Squaring both sides gives $x + 7 = 16$, so $x = 16 - 7$, and the check reads $\sqrt{9+7} = \sqrt{16} = 4$ ✓. **$x = 9$**

Try it: Solve $\sqrt{x+5} = 6$.

check: $x = 31$

(!) Watch out: perfect squares make the split look harmless — $\sqrt{36+64}$ “gives” $6+8 = 14$ so smoothly that nobody checks. Test any splitting step on numbers you can evaluate both ways; one counterexample settles it.